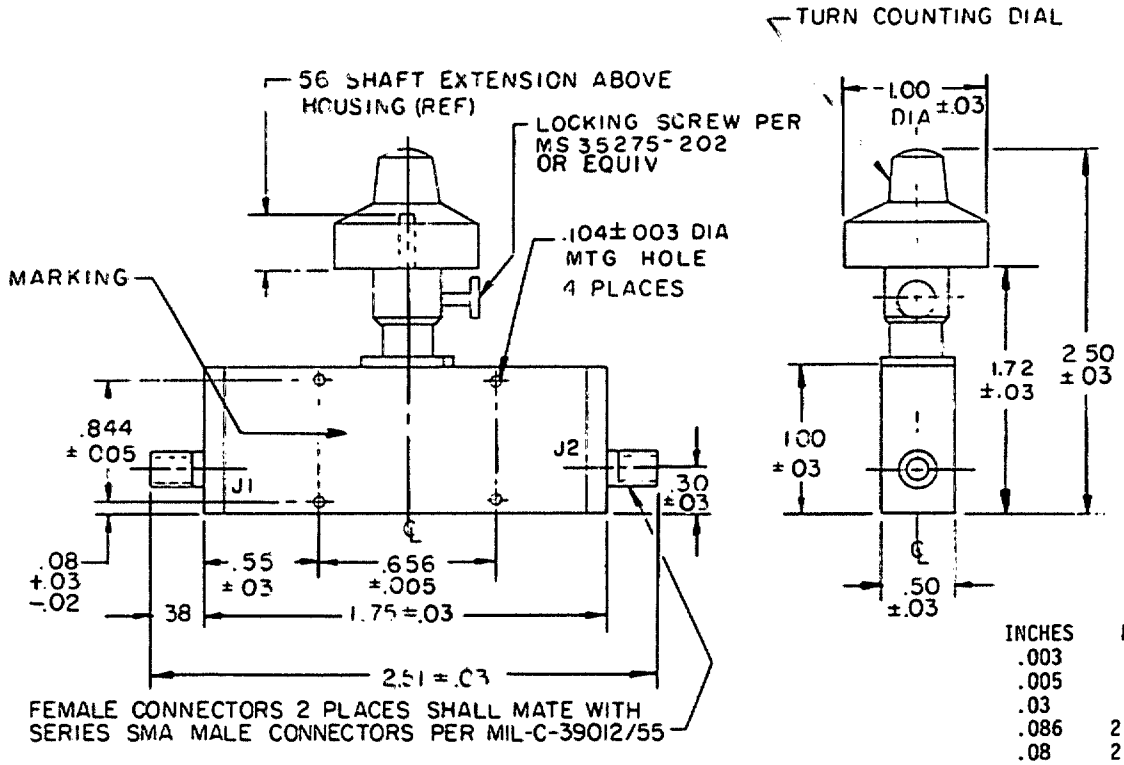
MILITARY SPECIFICATION SHEET

ATTENUATORS, VARIABLE (COAXIAL),
CLASS 5 (SMA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the attenuators described herein shall consist of this document and the latest issue of Specification MIL-A-24215.



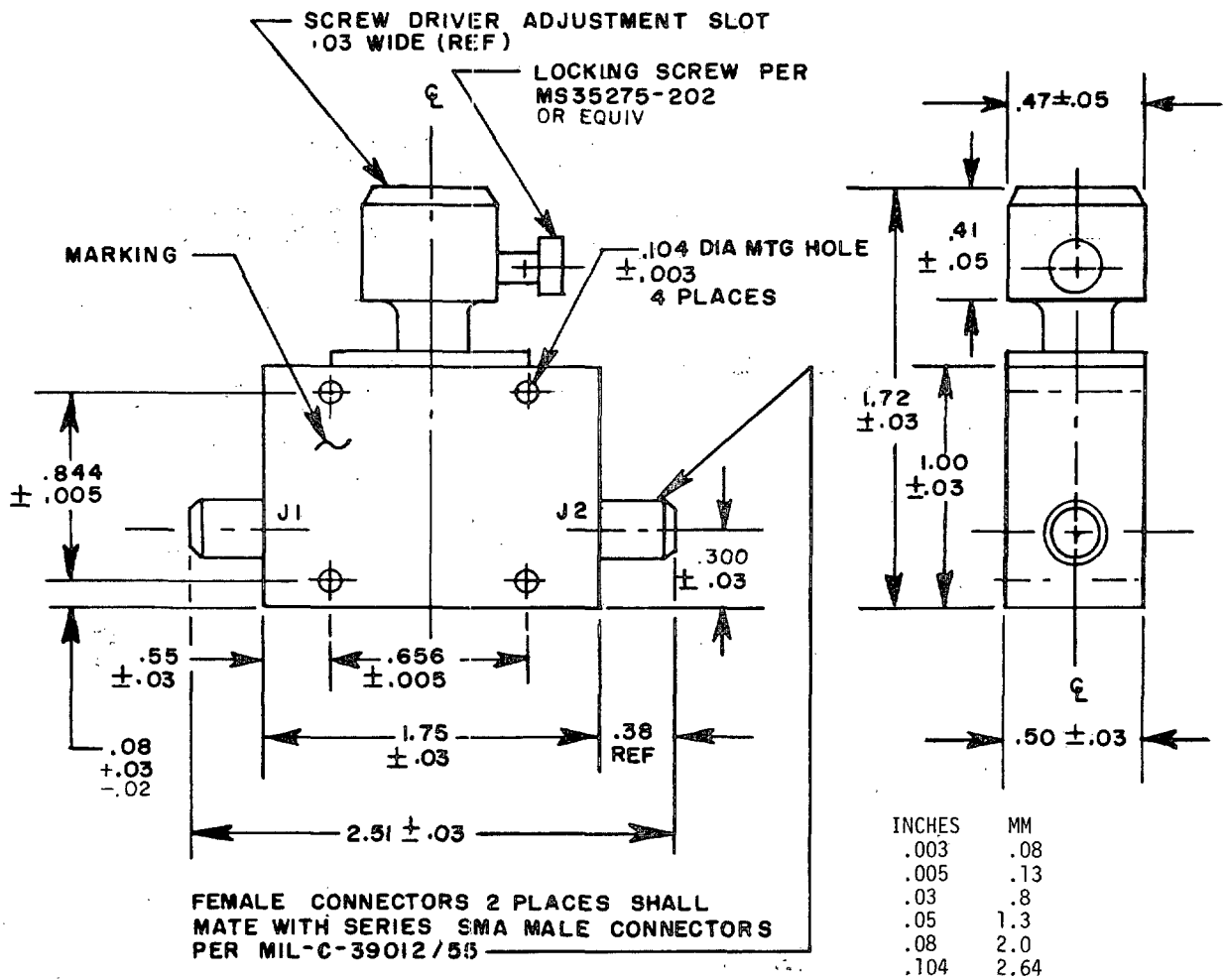
INCHES	MM
.003	.08
.005	.13
.03	.8
.086	2.18
.08	2.0
.104	2.64
.19	4.8
.30	7.6
.38	9.7
.50	12.7
.55	14.0
.656	16.66
.844	21.44
1.00	25.4
1.72	43.7
2.50	63.5
2.51	63.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1. Dimensions and configuration for part number M24215/9-001.

FSC 5985

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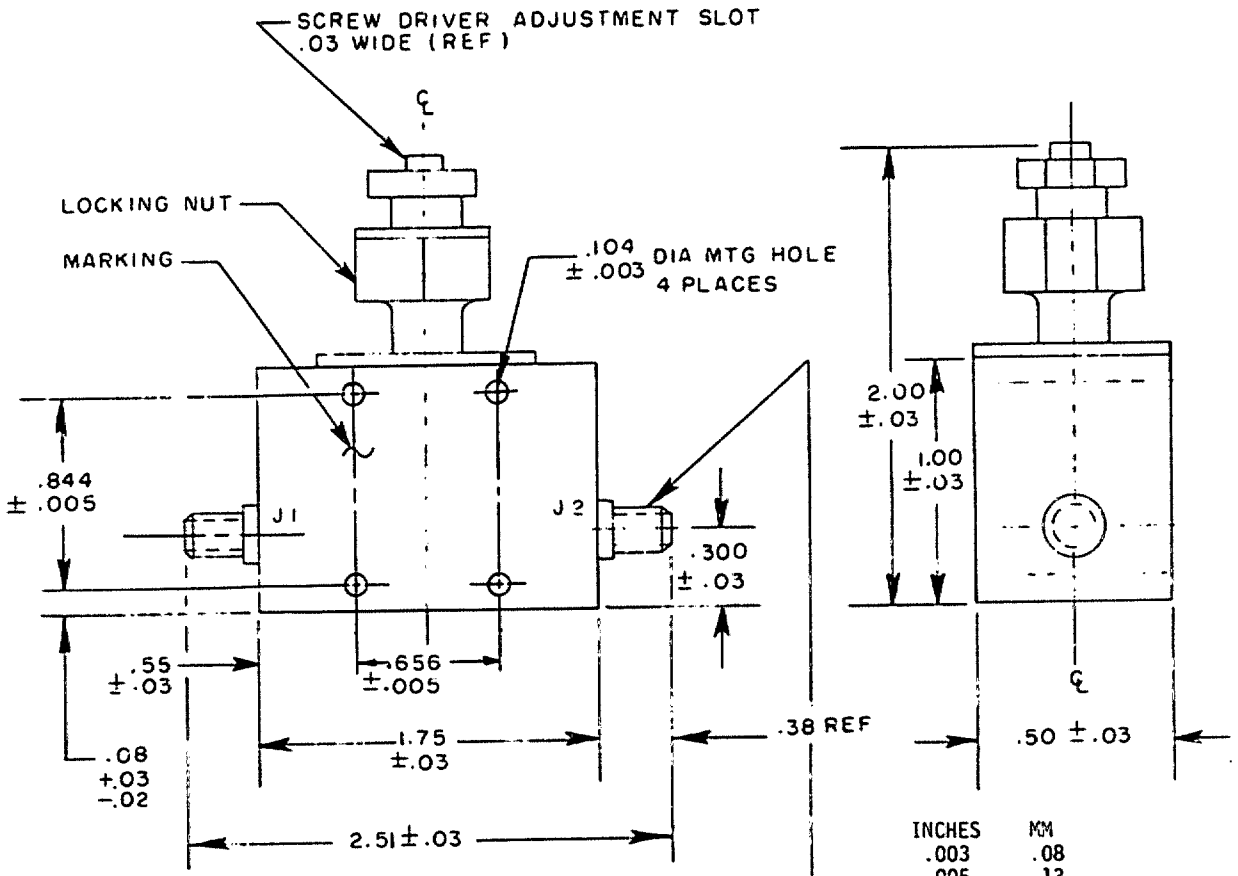


INCHES	MM
.003	.08
.005	.13
.03	.8
.05	1.3
.08	2.0
.104	2.64
.30	7.6
.38	9.7
.41	10.4
.47	12.0
.50	12.7
.55	14.0
.656	16.66
.844	21.44
1.00	25.4
1.72	43.7
1.75	44.4
2.51	63.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 2. Dimensions and configuration for part number M24215/9-002.

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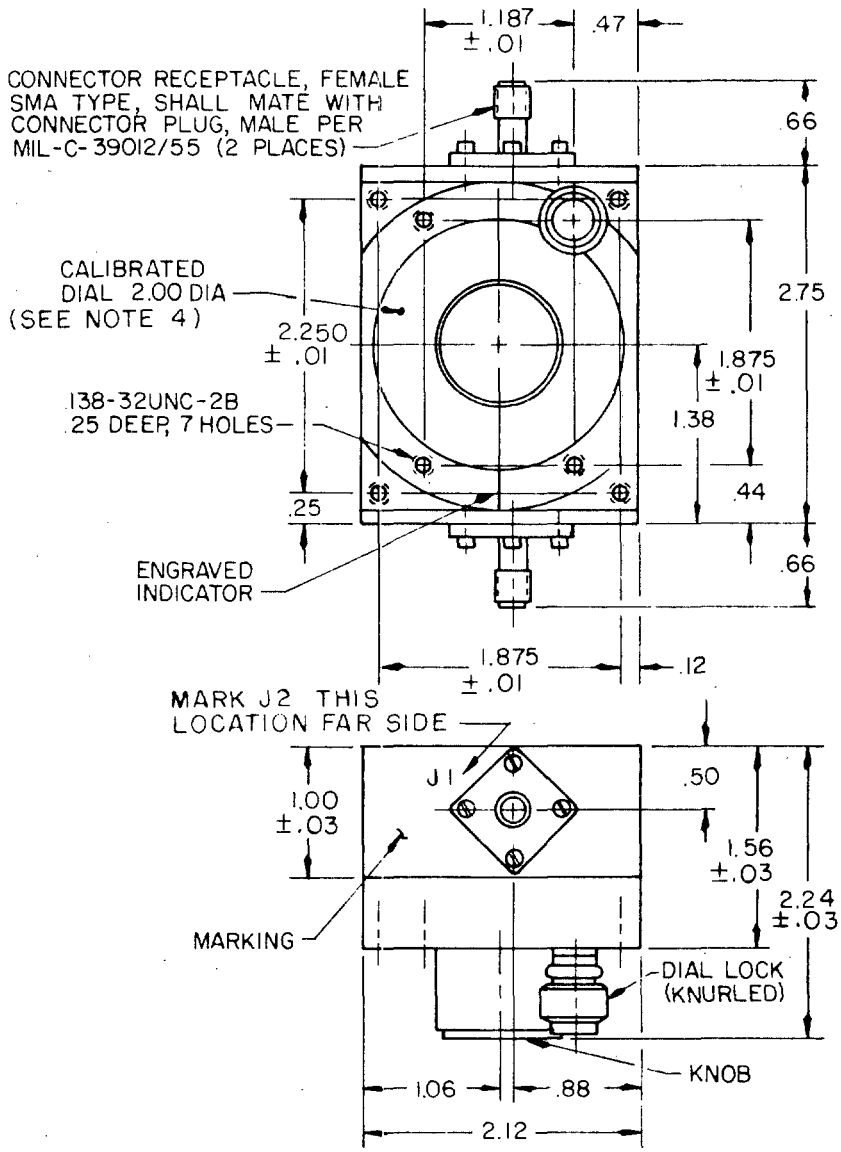


INCHES	MM
.003	.08
.005	.13
.03	.8
.08	2.0
.104	2.64
.30	7.6
.38	9.7
.50	12.7
.55	14.0
.656	16.66
.844	21.44
1.00	25.4
1.75	44.4
2.00	50.8
2.51	63.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 3. Dimensions and configuration for part number M24215/9-003.

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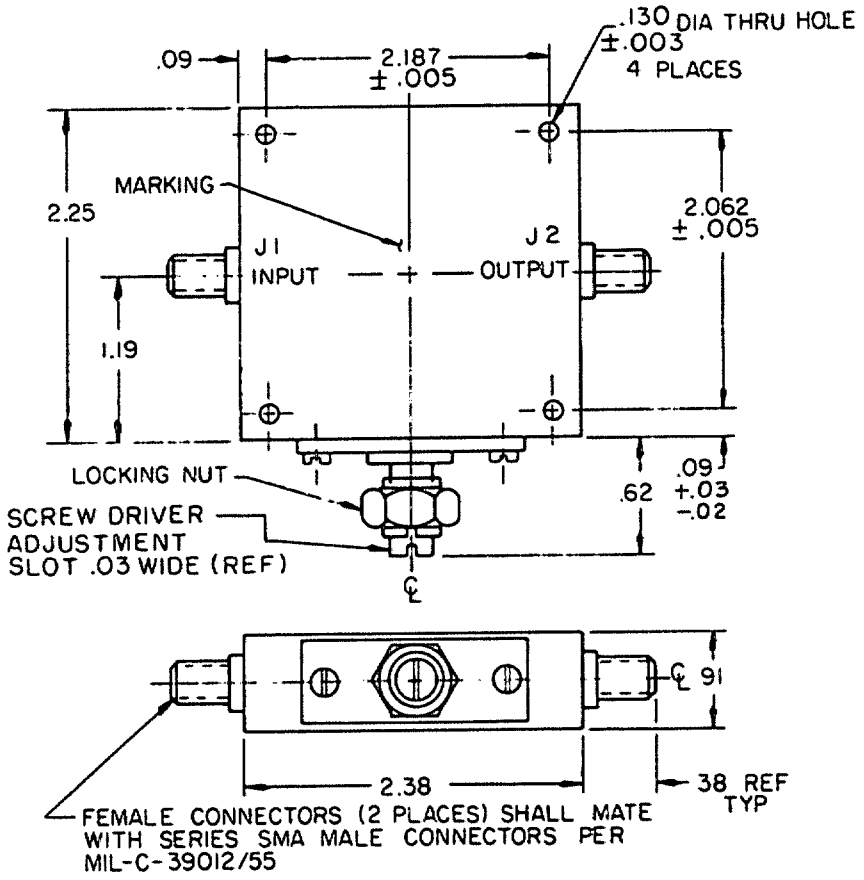


INCHES	MM
.01	.3
.03	.8
.12	3.0
.25	6.4
.44	11.2
.47	11.9
.50	12.7
.66	16.8
.88	22.4
1.00	25.4
1.06	26.9
1.56	39.6
1.187	30.15
1.875	47.62
2.24	56.9
2.250	57.15

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.02 (.5 mm).
4. Dial shall be marked in one dB steps to the 10 dB level, and in two dB steps through the 30 dB level.

FIGURE 4. Dimensions and configuration for part number M24215/9-004.

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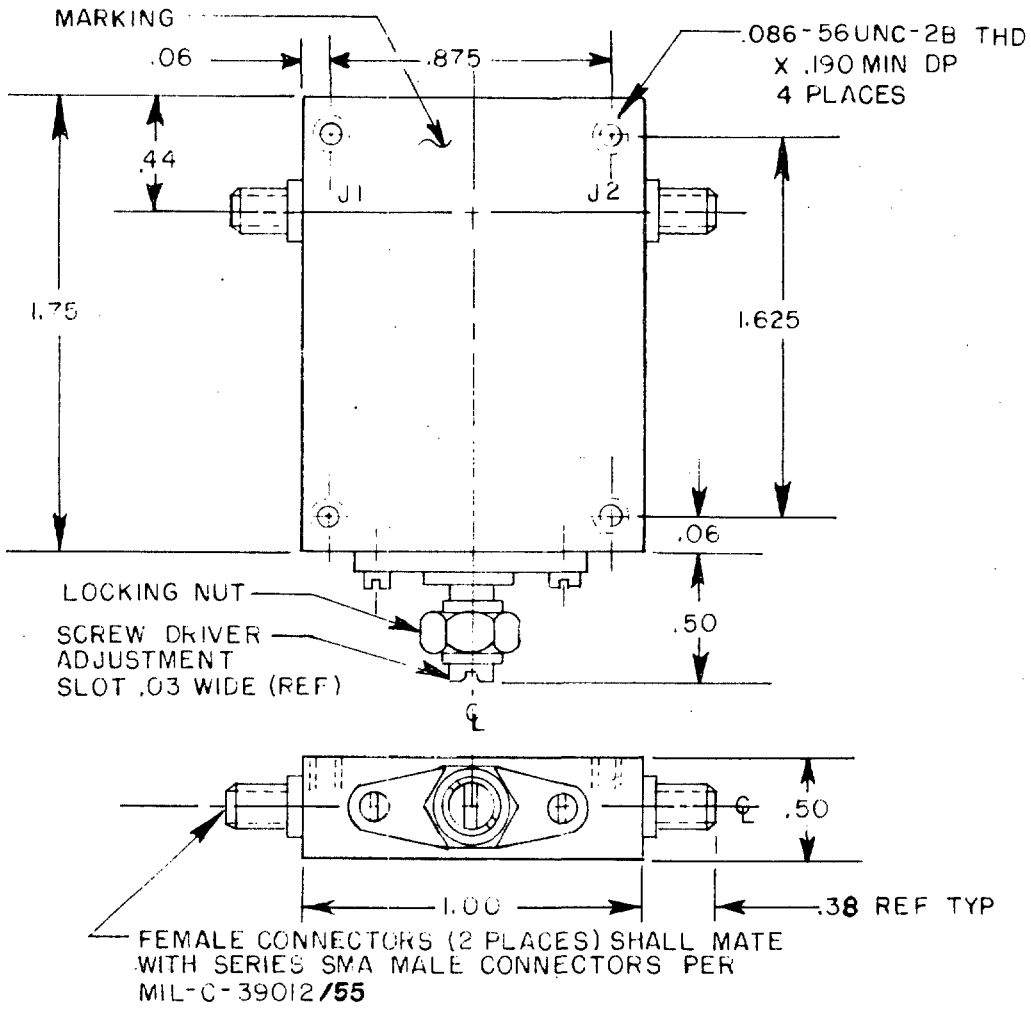


INCHES	MM
.003	.08
.005	.13
.09	2.3
.127	3.23
.133	3.38
.38	9.7
.62	15.7
.91	23.1
1.19	30.2
2.062	52.37
2.187	55.55
2.25	57.2
2.38	60.5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.03 (.8 mm).

FIGURE 5. Dimensions and configuration for part number M24215/9-005.

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INCHES	MM
.06	1.5
.086	2.18
.190	4.83
.38	9.7
.44	11.2
.50	12.7
.875	22.22
1.00	25.4
1.625	41.28
1.75	44.4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.03 (.8 mm).

FIGURE 6. Dimensions and configuration for part number M24215/9-006.

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REQUIREMENTS:

Dimensions and configuration: See figures 1, 2, 3, 4, 5, and 6.

Nominal impedance: 50 ohms.

Final finish: Flat black lacquer, color 37038 per FED-STD-595.

Electrical requirements: See table I.

Part number: See table I.

TABLE I. Electrical requirements.

Part number M24215/9-	Figure	Freq. range (GHz)	Atten. range (dB) 3/	VSWR (max)	Insertion loss (max at min. atten. setting) (dB)	Calibration accuracy (dB)	Frequency sensitivity (dB)	Min input (watts)		Nominal operating temp. range (°C)	Weight (max)		
								av.	pk.		lbs	(g)	oz
001	1	2-8	0-15 at 2 GHz 0-20 at 4-8 GHz	1.50:1	0.5	N/A	±1	2	---	-45 - +110 (See notes 1 and 2)	.16	(73)	(2.5)
002	2	2-8	0-15 at 2 GHz 0-20 at 4-8 GHz	1.50:1	0.5	N/A	±1	2	---	-45 - +110 (See notes 1 and 2)	.16	(73)	(2.5)
003	3	7-11	0-30	1.50:1	0.5	N/A	±1.5	2	---	-45 - +110 (See notes 1 and 2)	.16	(73)	(2.5)
004	4	9.0-9.2	0-30	1.50:1	0.7	±0.5	±0.3	2	200	-18 - +70	1	(454)	(16)
005	5	4.0-18	0-15	1.50:1	1.0	N/A	±2	5	3000	-30 - +105	.5	(227)	(8)
006	6	11-17	0-20	1.60:1	0.5	N/A	±2	5	3000	-30 - +105	.16	(73)	(2.5)

NOTES:
1. Non-operating ambient temperature range: -60°C to +110°C.
2. Temperature coefficient: 1.0 dB maximum change over the temperature range -45°C to +55°C at any attenuation setting.
3. Continuously variable.

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NOTE: Attenuators covered by this specification sheet are substituted for the following manufacturer's part number. This information in no way implies that the manufacturer's part is suitable as a substitute for the military part number.

Part number M24215/9	Manufacturer's code	Manufacturer's part number	FSN 5985
001	05869	723786-1	
001	12457	AUM-15A/906232	
002	05869	723786-2	
002	97942	580R580H02	136-4016
002	12457	AUM-15/802263	136-4016
002	12457	AUM-15A/11725	
002	12457	AUM-15/811054	460-5574
003	05869	721773	
003	12457	AUM-15H/13384	
004	49956	584073-1	
004	12457	AU-26/14844	
005	05869	721774-1	
005	99899	23702	
005	99899	4799	
006	05869	721776-1	
006	99899	23698	

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, SL, AR
Air Force - 11, 80
DLA - ES

User activities:
Army - SM
Navy - MC, AS, OS, SH

Preparing activity:
Navy - EC

Agent:
DLA - ES

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DOCUMENT IDENTIFIER AND TITLE

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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